SAI MADHUKAR K

Final-Year B.Tech. Robotics and Mechatronics Engineering
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Professional Summary

Final Year student pursuing B.Tech in Robotics and Mechatronics Engineering; having obtained a Diploma in Tool & Digital Manufacturing. Has designed and fabricated an injection mould, designed and tested a companion robot, and is now working on a vision-guided UAV using ROS 2 framework. Uses Fusion 360 and CATIA V5 in combination with Python, embedded hardware integration, and functional testing. Has undergone training in Drone Technology and Industrial Applied IoT from L&T EduTech.

Technical Skills

Mechanical Design: Fusion 360, CATIA V5, 3D part modelling, assembly modelling, engineering drawings, drafting, GD&T, tolerances.

Manufacturing & Testing: Injection mould design and fabrication, mechanical assembly, prototype integration, functional testing.

Programming & Robotics: Python, ROS 2, Linux/Ubuntu, OpenCV, YOLOv8n, computer vision, autonomous navigation development.

Hardware: Raspberry Pi 4, Arduino, ESP32, servo actuators, DC motors, motor drivers, sensors, Pixhawk 6X.

Simulation & UAV Tools: Gazebo, RViz, MAVLink, MAVSDK, QGroundControl.

Project Experience

Companion Robot - Mechanical Design & Hardware Integration

Fusion 360; Raspberry Pi; servo actuators; DC motors

- Designed a mobile companion robot in Fusion 360 with a four wheel driven system, articulating arm,display and joystick controls.
- Integrated mechanical and embedded systems with Raspberry Pi, servo actuators, DC motors, motor drivers, and sensors.
- Developed and tested voice assistance features as well as an emoji face on the actual robot prototype,showing the audio and visual interaction capabilities.
- Servo control was integrated and confirmed to work on the prototype.
- Presentated the research for the companion robot at I-4AM'26, IISc Bengaluru

Injection Mould Design & Manufacturing

Fusion 360; engineering drawings; tolerances; fabrication; assembly

- Performed the entire process of taking an injection mould from concept design to manufacture including part creation, mould assembly,fabrication and assembly.
- Designed 3D Models and assembly designs in Fusion 360 to create the mould and its assembly.
- Prepared engineering drawings with dimensions and tolerances for fabrication, connecting CAD work with the requirements of physical assembly.

Vision-Guided Autonomous UAV

2026-Present | Ongoing project

- Developing a vision-guided UAV for aerial crowd monitoring using ROS 2, Python, Raspberry Pi 4, and Pixhawk 6X.
- Working on camera integration and ROS 2 image processing using OpenCV and YOLOv8n for crowd and obstacle detection.
- Integrating MAVLink, MAVSDK, and QGroundControl for flight-control communication; developing vision-based adaptive navigation as part of the ongoing system.

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Mechanical Design | Robotics | Manufacturing

Internship Experience

Engineering Design Intern

Government Tool Room & Training Centre (GTTC), Bengaluru | May 2026

- Used CATIA V5 for mechanical part modelling, assembly design, and drafting during engineering design internship work.
- Interpreted engineering drawings and gained exposure to dimensional accuracy and manufacturing-oriented design practices.

Mechanical Engineering Intern

Universal Engineering, Tiruppur | April-May 2025

- Participated in hands-on mechanical design and programming activities, gaining exposure to industrial engineering and manufacturing processes.

Education

B.Tech. in Robotics and Mechatronics Engineering

CHRIST (Deemed to be University), Bengaluru | 2024-2027

Lateral Entry; expected graduation: 2027

Overall: 68.42% | Semesters 3-6 GPA average: 3.13/4.00

Diploma in Tool & Digital Manufacturing

Nettur Technical Training Foundation | 2021-2024

Overall: 90.2%

Leadership & Activities

Project Lead - IEEE Robotics and Automation Society

2025-2026

- Coordinated robotics and automation project activities, including technical development, task allocation, and team execution.
- Supported project documentation and technical communication across team activities.

MSME IDEA HACKATHON 6.0: Developed and presented an innovation concept as part of a student team.

Professional Training

Drone Technology - L&T EduTech

Industrial Applied IoT - L&T EduTech

Robotics and IoT, Mindentious - April-May 2026

Practical training covering robotics, sensor integration, IoT concepts, and connected robotic systems.

Drone Mechanics, Mindentious - March-April 2026

Training covering drone mechanical systems, components, and mechanisms.

Certifications

NOCN Level 5 Certification

NSDC Level 3 & Level 4 Certifications